

Fachry Dennis Herald

Jakarta, Indonesia | +62 882 1185 4646

fd.heraldi@gmail.com

linkedin.com/in/dennisheraldi

dennisheraldi.github.io

github.com/dennisheraldi

Software engineer with experience integrating different microservices into one connected solution. Quick to adapt to new technology, focused on building reliable systems that can recover from failures, and committed to clear documentation and knowledge sharing.

Work Experience

CSG International

Software Developer Engineer II

Jakarta

April 2026 - Present

CSG is a global provider of customer experience, billing, and payment solutions for the telecom industry.

- As an early adopter of AI tooling at the company, built an **AI Email Interpreter** in Java with **LangChain** on a self-hosted **Ollama qwen3-8b** model that converts multilingual client emails (**validated in English and Mandarin**) into structured order-status payloads, using **example prompts** and falling back to **manual review** when confidence is low; **delivered in 2 days for a client hackathon** and selected by pre-sales to **demonstrate order-to-inbox capability at MWC Barcelona 2026**.
- Built an **end-to-end AI agent** that turns **unstructured RFP documents** into **fully-configured Quote & Order records**, on top of an **open-source agent-orchestration framework** (deterministic **node-graph workflow**, **tool-calling**, and **MCP integration** with a bundled Java service); it reads each document with a **vision LLM**, qualifies every site against multiple providers, and assembles a multi-product quote with per-site configuration — **compressing hours of manual catalog lookup, site qualification, and quote building into a few-minute run**, demonstrated live on the **Catalyst** showcase environment.
- Won 2nd place at the internal AI competition** with an **AI-Assisted API Transformation engine** that converts natural-language instructions into transformation scripts, **automating an API mapping step that previously required manual scripting** during customer onboarding.
- Built an **AI Analytics PoC** that exposes the production database through a **chat interface backed by MCP**, giving **business users self-service access to operational data** without writing SQL.
- Lead developer for Banking & Financial Solutions (BFS)**: prepared **10+ international demos at banking events including Sibos**, owned demo environment microservices, and served as **first-line on-call**; also built supporting AI tools (a **PDF reader for financial statements** and an **MCP server that coordinates API calls**) that **cut demo preparation time for later events**.

Software Developer Engineer I

January 2025 - April 2026

- Engineered an engine for intent-driven network API transformations** as the MVP of a Catalyst project, **contributing to the company winning the TMForum DTW 2025 award** with Ericsson.
- Reached zero transaction failures** by adding end-to-end recovery with **idempotency, process-engine auto-retry, and stronger internal error handling**, keeping critical business operations reliable.
- Modernized 50 customized XML product specifications** and Java microservices, refining core business processes to make it easier for clients to configure the product.

Paper.id

Software Engineer

Jakarta

November 2023 - December 2024

Paper.id is a platform providing invoicing, payment, and financing solutions for businesses.

- Cut message template creation from hours to under a minute** by integrating third-party SMS and WhatsApp broadcast services with **concurrent Goroutines processing**.
- Built a unified payment engine** with a generalized design that handles **different bank behaviors and merchant-specific requirements**, making it easier to integrate with multiple parties while **keeping the system consistent across financial partnerships**.
- Reached a 99% transaction success rate** by implementing QRIS, direct transfer, virtual account, and cross-border remittance flows that follow **Indonesia's Central Bank standards**, contributing **4 billion IDR TPV** in a month.
- Hardened the system** by adding **dynamic table partitioning and automated transaction reconciliation** through RabbitMQ scheduling and async callbacks, **improving query performance by 20%** and making reconciliation reports run instantly.

Suitmedia

Software Engineer Intern

Bandung

May 2023 - September 2023

- Built a **Laravel-based CMS** with **35 dynamic menus and 50 API endpoints** from Figma prototypes, with database schemas designed for **smooth integration with the front-end**.
- Redesigned database structures** and implemented **dynamic content management with static page integration**, improving **query performance by 30%** and enhancing the user experience across both dynamic and static pages.

Synpulse

Software Engineer Intern

Jakarta

October 2022 - December 2022

- Authored 20 functional requirement specifications** and mapped API specifications for a **core banking system migration**, creating system flowcharts that kept the **Banking Client's Avaloq transition project on schedule**.
- Removed manual daily data processing** by writing automated **Excel-to-Oracle shell scripts**, cutting reconciliation report processing **from 15 minutes to near-instant**.

Education

Institut Teknologi Bandung

Bachelor of Engineering in Informatics

August 2020 - October 2024

CGPA: 3.76/4.00 (Cumlaude)

- Relevant Coursework**: Algorithm and Data Structure, Software Project Management, Database Management, Platform-based Application Development, Parallel and Distributed Systems, Advanced Machine Learning

Awards & Recognition

Sarcasm Detection Using Fine-tuned Llama 2

International Conference on Advanced Informatics (ICAICTA 2024)

Singapore

September 2024

- Best Paper Award**. Fine-tuned **Llama 2** for sarcasm detection on a dataset of **21,000+ samples**, **beating previous benchmarks by 8%** through prompt engineering and parameter-efficient fine-tuning trade-offs that help the model understand nuanced language better.

Technical Skills

- Programming Languages**: Java, Golang, PHP, Python, JavaScript, Shell Scripting
- Frameworks & Libraries**: Laravel, React, ViteJS, NodeJS, Fastify, JAX-WS

- **Databases:** PostgreSQL, MySQL, ArangoDB, Oracle Database
- **Tools & Infrastructure:** RabbitMQ, Kafka, OpenTelemetry, Vault, Consul, Docker, Jenkins, Google Cloud Platform, AWS
- **AI / ML:** LangChain, Ollama, OpenRouter, Llama 2, Qwen, Model Context Protocol (MCP), Prompt Engineering, LLM Evaluation